Hussain Singapuri

AI AUTOMATION ENGINEER (MERN, LLM)

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PROFESSIONAL SUMMARY

AI Automation Engineer with 3 years of professional full-stack experience at a US-based medical-device company. Builds **AI agents**, **LLM** integrations (Claude and Gemini APIs, function calling, structured JSON outputs), **RAG** pipelines with embeddings and vector search, and **workflow automation** with webhooks and CRM integrations. Works in **JavaScript, TypeScript, React and Node.js** and uses **Claude Code** daily. Claude Certified Developer.

CORE COMPETENCIES

LLM API Integration | Function Calling & Structured JSON | Prompt Engineering | RAG, Embeddings & Vector Search | AI Agents & Workflow Automation | REST APIs & Webhooks | React, Node.js & TypeScript | AI Coding Assistants (Claude Code)

WORK EXPERIENCE

NeuroNetworks AI Ed

2026 - Present

CTO & Principal Course Architect

Pune, India

- Built and run the website neuronetwrks.com with a Python static-site generator deployed on Vercel, including programmatic SEO
- Lead technology for an Indian AI and tech education brand (Neuronetworks AI Ed Private Limited) offering programmes for students after Class 10
- Architected the curriculum: a 1-year AI Foundation course and five 3-year specialisations (AI/ML Engineering, Software Development, Data Science & AI, Product & UX / AI Design, Cybersecurity)
- Designing Neuromind Practical AI Lab, a browser-based IDE and sandbox with an AI tutor: phased spec, server sizing, engineering team plan and CAPEX/OPEX model

US-based medical-device company

Mar 2023 - Sep 2026

Full Stack Developer

Remote

- Built and maintained full-stack web applications for the company
- Built **AI agents** for internal use
- Built several apps, including a **CRM**
- Performed penetration / security testing on 5+ of the company's mobile apps, all live on the Google Play Store

PROJECTS

site-to-agent

Open Source

Turns any website into a grounded chat agent: crawler, hybrid BM25 + vector retrieval with RRF, cited answers, SSRF-hardened fetching and an eval harness.

Next.js, TypeScript, pgvector

multi-agent-research-crew

Open Source

Planner, parallel researcher, writer and verifier agents on a custom DAG orchestrator with retries, timeouts, token budgets and claim-level citation checking.

Python, asyncio, Claude

ai-lead-receptionist Open Source

AI receptionist for home-service businesses: missed-call text-back, SMS and web chat that qualifies leads and books jobs, owner dashboard.

Next.js, TypeScript, Twilio, Claude, Supabase

PageForge Portfolio project

Account-based marketing page generator: AI drafts only from approved facts and the account's public pages, a deterministic validator flags unsupported numbers, a reviewer signs off claim by claim, engagement flows to analytics and the CRM.

Next.js 16, React 19, Tailwind v4, Sanity, Supabase, Gemini, PostHog, HubSpot

Orbital Surplus Portfolio project, fictional brand

Full-stack store with race-safe stock holds in one transaction, idempotent payment confirmation via signed Stripe webhook (test mode), customer accounts, admin with partial refunds; 9 end-to-end API tests.

three.js, GSAP, WebXR, Node 20, Postgres (Neon), Stripe, Resend

Personal portfolio with full-stack backend Portfolio project

Lead pipeline with rate-limited enquiry form, booking with free intro calls and paid Stripe sessions (slot holds, signed webhooks), HMAC-signed CRM events, nonce-based CSP, 10 end-to-end API tests.

Vanilla ES modules, GSAP, Node 20, Postgres (Neon), Stripe, Resend

More projects

nanoserve, lead-intel-engine, contractor-ai-ops-dashboard, Nightjar & Co., Lumen, Kilnwood Roasters, Cairnmoor, Lowlight Video Club, Safey Inventory, TripSync

EDUCATION

Bachelor's, Computer Science (AI & Machine Learning) Vishwakarma Institute of Technology, Pune 2022-2025

Pune

CERTIFICATIONS

Claude Certified Developer	Anthropic, Foundations (CCDV-F)
Claude Certified Orchestrator	Anthropic, Foundations (CCAO-F)
Generative AI Leader	Google Cloud
Claude Code in Action	Anthropic
Claude Code 101	Anthropic

SKILLS

Back end: Node.js, FastAPI, asyncio, REST APIs, webhooks (HMAC signing), auth (scrypt, sessions, throttling), Stripe Checkout and refunds, Resend

Front end: React 19, Next.js, TypeScript, JavaScript, Tailwind CSS, shadcn/ui, TanStack Start / Query, GSAP, Lenis, three.js, WebGL, Canvas 2D, responsive and accessible UI, motion and interaction design

AI / LLM: Claude API, Gemini API, AI agents, multi-agent orchestration, tool use / function calling, RAG (hybrid BM25 + vector, RRF), evals, prompt-injection defence, LLM inference serving, Claude Code and AI coding agents

Languages: TypeScript, JavaScript, Python, SQL, Dart, HTML, CSS

Data: PostgreSQL, Neon, Supabase (RLS, Edge Functions), PGlite, pgvector, SQLite, Sanity CMS

DevOps: Git, GitHub Actions, Docker, Vercel

Security: mobile app penetration testing, CSP and security headers, SSRF hardening, input validation

Automation & integrations: workflow automation, CRM integrations (HubSpot), PostHog, Twilio SMS, web crawling / scraping

Mobile: Flutter, Riverpod, Firebase **Design:** UI/UX design, design systems, technical SEO, JSON-LD structured data, programmatic SEO